Assignment – 2.9

Introduction to Network Protocols

Q - 1. Use Wireshark to capture network traffic while you load www.spotify.com in your browser, then identify and note down at least two DNS packets and two HTTP or HTTPS packets in the capture.

- **Identify DNS packets:**

Packet	Protocol	Description
1	DNS	Standard query for www.spotify.com
2	DNS	Standard query response containing the IP address of Spotify

- **Identify HTTPS packets:**

Packet	Protocol	Description
1	TLS	Client Hello – Browser starts a secure connection with Spotify.
2	TLS	Server Hello – Spotify's server responds and negotiates security settings.

Q - 2. Simulate a DNS lookup for www.flipkart.com using the nslookup command in your terminal and write down the IP address returned along with a brief explanation of what this result means.

- **DNS Lookup Using nslookup:**

  Name: www.flipkart.com

  Address: 163.53.78.128

- **Explanation:**

  The nslookup command is used to find the IP address of a website.

  - Website name: www.flipkart.com
  - IP address: 163.53.78.128 (sample result)
  - This IP address identifies the server that helps our computer connect to Flipkart.

Q - 3. Explain the difference between HTTP and HTTPS by listing 3 key differences, and give a real-world example of an app or website you use that relies on HTTPS for secure communication.

- **Difference Between HTTP and HTTPS:**

  HTTP and HTTPS are protocols used to transfer data between a web browser and a website.

- **Three key differences:**

No.	HTTP	HTTPS
1	Data is not encrypted.	Data is encrypted and secure.
2	Less secure.	More secure.
3	Uses port 80.	Uses port 443.

- **Real-world example:**

  Google, Flipkart, and Spotify use HTTPS to protect user information such as passwords, login details, and payment information.

Q - 4. Open Wireshark and filter for DNS protocol traffic. Try searching for a new website (one you haven't visited recently) in your browser and describe the sequence of DNS queries and responses you observe. Hint: Look for packets with 'Standard query' and 'Standard query response' in the Info column.

- **DNS Traffic Capture Using Wireshark:**
- **DNS Query and Response:**

1. **Standard Query:** The computer sends a DNS query to find the IP address of www.amazon.in.
2. **Standard Query Response:** The DNS server replies with the IP address of the website.
3. **Connection:** The browser uses the received IP address to connect to the website.

- **Result:**

Wireshark shows DNS query and response packets. This helps us understand how DNS converts a website name into an IP address so that the browser can open the website.

<u>Q - 5.</u> Pick any one protocol from DNS, DHCP, FTP, or SNMP (other than HTTP/HTTPS) and write a short scenario describing how it would be used in a real-world app or service you use (for example, how DHCP helps your phone connect to WiFi when you open Instagram at a cafe).

- **<u>Protocol: DHCP (Dynamic Host Configuration Protocol)</u>**
- **<u>Real-world scenario:</u>**
  - When I connect my phone to Wi-Fi at a cafe and open Instagram, DHCP automatically assigns my phone an IP address, subnet mask, default gateway, and DNS server.
  - This allows my phone to connect to the internet and use Instagram without manually entering network settings.